

PRODUCTION ROADMAP

TradePilot v6.1

From a 6-engine paper trading laptop to a SEBI-compliant, multi-broker, optionally autonomous production system

36-40

Weeks to public launch

5

Statistical decision
gates

80%

Phase 1 complete today

Wednesday, 29 April 2026

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Reconciles April 12 v6 master plan with shipped Track A + Fix #1

Naming clarification

Two things are called "v6" right now. This doc fixes that.

Name	What it is	Lifecycle
v6 (the engine)	Paper-trading experiment shipped 2026-04-28 EOD. "v4 raw signals + Track A bolt-on." Lives in <code>scripts/v6-paper-trade.py</code> . One of 6 engines running on the laptop.	Experiment. Will be retired or promoted at the May-25 statistical gate.
v6.1 (this roadmap)	The full production system from the April 12 master plan, refreshed with everything we've learned and shipped since then. The cloud-hosted, multi-broker, optionally autonomous TradePilot.	Roadmap. 9-12 months from today to public launch.

From here on: "v6" always means the paper-trading engine. "v6.1" always means the production system.

Where we actually are tonight

A clean snapshot before we plan forward.

Engines in observation (6 active)

Engine	Role	Edge being tested
v4	Control. Raw v4 composite scorer, LONG-only.	Baseline. 04-28 made +Rs 26,179 (71% WR).
v5 (Fix #1)	v5 wrapper + tonight's Fix #1.	Does Fix #1 close the gap to v4?
v5_classic	Baseline v5 with no Track A.	Confirms Track A's contribution.
v5_6	v5 + Darvas Box overlay.	Independent variant.
v5_7	v5 + Intraday Box overlay.	Independent variant.
v6	v4 raw + Track A bolt-on.	Cleanest test of Track A's marginal value.

What's already shipped

Layer	Status
Multi-engine paper trading framework	Live, 6 engines, daily reports
Track A defensive layer (cost modeling, RE-ARM, SHORT_BLOCK, FLAT_EXIT)	Shipped 2026-04-27
Fix #1 (SHORT requires absolute weakness)	Shipped 2026-04-28
ML staging + 4- gate promotion system	Shipped — no model auto-promotes to production
Crash watchdog	Shipped — auto-restart, market-hours-aware
Telegram digest (2-hour cadence)	Shipped
Laptop heartbeat (15- min)	Shipped
Auto-stop-EOD (15:35)	Shipped
Daily watchdog learning insights	Shipped
LaunchAgent for Monday weekly tracker	Shipped

Layer	Status
LaunchAgent for tomorrow's EOD comparison	Shipped tonight
Statistical gate doc (deflated Sharpe + 95% CI)	Live in <code>docs/IMPLEMENTATION_BRIEF_2026-04-27.md</code>
Threat surface analysis (1,597 lines)	Live in <code>docs/security/TRADEPILOT_V6_LIVE_TRADING_THREAT_ANALYSIS.md</code>
Competitive analysis (Zerodha, Groww, Streak, Sensibull, etc.)	Live in <code>docs/research/competitive-analysis.md</code>

Where the original April 12 plan put us

"Phase 1: Foundation, Weeks 1-4. Make it work."

Reality: We are at session ~14 of 20-session paper validation. Track A (which wasn't in the original plan) ate the last 4 weeks and was correct to do so. **Phase 1 is ~80% complete.** Phases 2, 3, 4 are unchanged from the original plan.

The four-phase staircase (v6.1, refreshed)

Original plan had 20 weeks. Reality says 36-40 weeks across 4 phases. Here's the honest version:

Phase 1 – Foundation ("make it work")

Goal: prove on paper that at least one engine produces statistically significant alpha.

Item	Original (Apr 12)	Refreshed (Apr 29)	Status
Paper trading framework	Build	Built (multi-engine)	Done
20-session validation	4 weeks	4-week observation window through 2026-05-25	In progress
ML model with proper training	Foundation	Done (LightGBM, best_iter ≥ 100 guardrail, atomic promote)	Done
Track A defensive layer	Not in plan	Shipped — 4 fixes	Done (added April 27)
Statistical gate (Sharpe + CI)	Not in plan	Lopez de Prado deflated Sharpe + 95% CI methodology	Methodology defined, awaiting data
Cross-asset signals (bonds, crude, USD/INR)	Foundation	Deferred to Phase 2	Not started
Shoonya account for historical data	Foundation	Deferred — not needed for current observation	Optional

Exit criteria for Phase 1 → Phase 2: At the May 25 gate, at least ONE engine must show a deflated Sharpe ≥ 1.0 with 95% CI lower bound > 0 over the 20-session window. If yes, that becomes our base engine. If no, we extend observation by 4 more weeks and revisit.

Time remaining in Phase 1: ~4 weeks (through May 25, 2026).

Phase 2 – Intelligence ("make it smart")

Goal: layer additional alpha sources on top of the winning Phase 1 engine.

Item	Original (Apr 12)	Refreshed (Apr 29)
LLM sentiment layer (Claude API for news)	Yes	Yes — but only if Phase 1 base engine alone falls below Sharpe 1.5
FII/DII institutional flow signals	Yes	Yes — public data, Rs 0 cost
Insider buying clusters from SEBI filings	Yes	Yes — public data
Pairs trading (cointegrated Nifty 200 pairs)	Yes	Yes — separate strategy pool
Options strategies (premium harvesting)	Yes	Defer to Phase 3 — too risky to layer with equity in early Phase 2
Multi-agent orchestrator	Yes	Soft-yes — only if monolithic loop becomes the bottleneck

Time: 8 weeks (May 26 → July 21, 2026).

Exit criteria for Phase 2 → Phase 3: Combined system (base engine + 2-3 alpha layers) shows Sharpe > 1.5 , max drawdown $< 12\%$, win rate $> 60\%$ on a fresh 30-session paper window.

Phase 3 – Execution ("make it real")

Goal: connect to a real broker, execute real orders with real money — but tiny size and hard kill switches.

Item	Original (Apr 12)	Refreshed (Apr 29)
Open Zerodha Kite Connect dev account	Yes	Yes — Rs 2,000/month, apply during Phase 2
Build Kite execution agent	Yes	Yes — separate microservice with audit log
Smart order routing (limit vs market)	Yes	Yes
Shadow mode (log orders, do not execute)	Yes — 2 weeks	Yes — extended to 4 weeks. This is the most important step in the entire plan.
SEBI algo registration via Kite	Yes — 2-4 weeks	OPEN — needs fresh SEBI April 2026 rules research. Different from RIA.
Kill switch + audit trail	Yes	Yes — daily, weekly, monthly loss caps with hard auto-halt
Move from laptop to AWS Mumbai	Yes (ap-south-1)	Yes — same VPC as Kite servers for sub-50ms latency
Live with 1 lot	Yes — 2 weeks	Yes — extended to 4 weeks. No exceptions.

Time: 12 weeks (July 22 → October 13, 2026).

Exit criteria for Phase 3 → Phase 4: 4 weeks live with 1 lot, no catastrophic incidents, real-money Sharpe within 1 standard deviation of paper Sharpe (this proves paper-to-live transfer worked).

Phase 4 – Scale ("make it bigger")

Goal: scale capital, add second-order features, prepare for public-facing offering.

Item	Original (Apr 12)	Refreshed (Apr 29)
Scale 1 lot → full personal capital	Yes	Yes — over 4-6 weeks, 25/50/75/100% ladder
Grafana real-time P&L dashboard	Yes	Yes — separate from paper-trading dashboard
Commodities (gold, crude)	Yes	Soft-yes — only if equity engine plateaus
Currency hedging (USD/INR)	Yes	Defer — adds complexity without clear edge
Tax optimization (STCG/LTCG awareness)	Yes	Yes — automated booking
Public API for subscribers	Yes	OPEN — depends on SEBI RIA license decision (separate workstream)

Time: 16 weeks (October 14, 2026 → February 2, 2027).

Decision gates (statistical, not vibes)

The gates are what stop us from "feeling like" the system is ready and shipping it anyway.

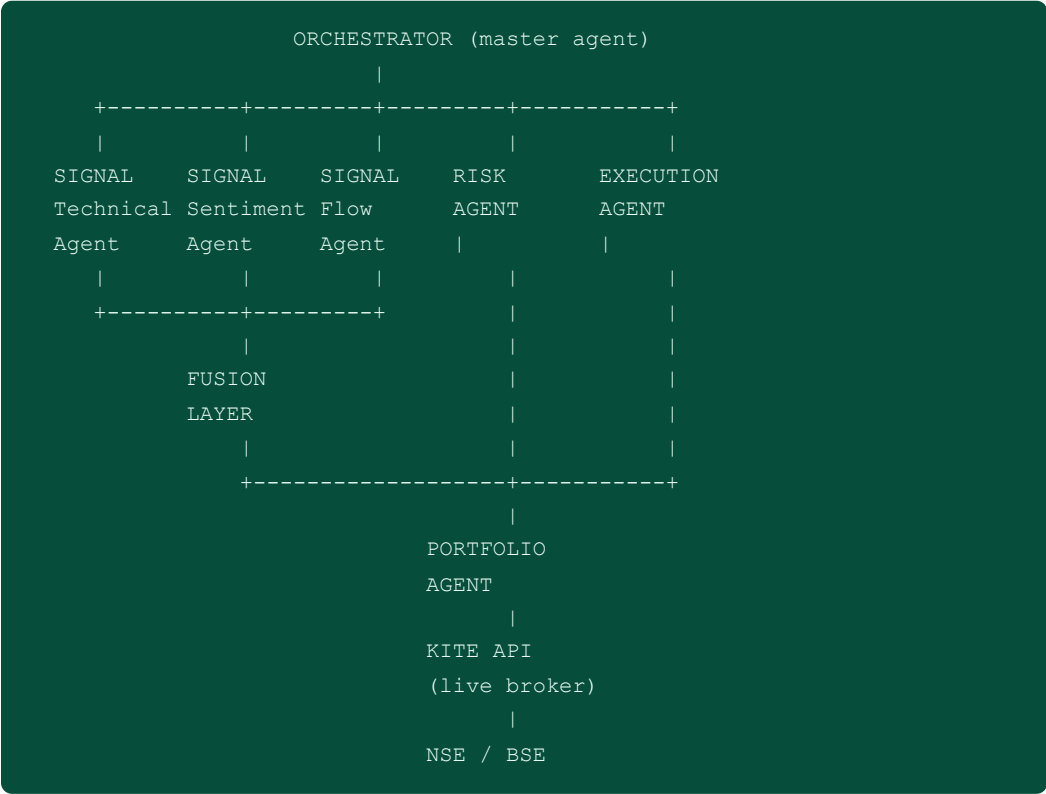
Gate	Date	What we measure	What passes
G1	2026-05-25	Phase 1 → Phase 2	At least one engine: deflated Sharpe ≥ 1.0 , 95% CI lower bound > 0 , 20+ trading days
G2	2026-07-21	Phase 2 → Phase 3	Combined system: Sharpe > 1.5 , max drawdown $< 12\%$, win rate $> 60\%$, 30+ trading days fresh post-Phase-1
G3	2026-08-31	Shadow mode → Live with 1 lot	4 weeks of shadow mode with zero divergence between predicted and would-have-been-actual fills
G4	2026-10-13	1 lot → Full personal capital	4 weeks live: real-money Sharpe within 1 SD of paper Sharpe; zero unauthorised orders; zero kill-switch overrides
G5	2027-Q1	Personal → Public	Determined separately by SEBI license + 6+ months of audited live track record

If any gate fails, we extend that phase. We do not skip gates.

Production architecture (v6.1)

Refreshed from the April 12 master plan. The core idea is unchanged: separate **signal generation** from **risk management** from **execution** from **portfolio construction**. Each is its own service, each is independently testable.

Service map



What each service does

Service	Job	What goes wrong if it fails
Orchestrator	Coordinate all agents, manage state, hold the global kill switch	Whole system halts safely (correct behaviour)
Technical signal agent	LightGBM composite scorer + Track A rules (this is v5/v6 today)	Lose primary alpha source — fall back to no-trade
Sentiment signal agent	Claude API news scoring (Phase 2)	Lose secondary alpha — degrade to technical-only
Flow signal agent	FII/DII + insider + delivery % (Phase 2)	Lose tertiary alpha — degrade gracefully
Fusion layer	Combine signals into a single decision per stock	Single point of failure — needs heavy testing
Risk agent	Regime detect, VIX size, kill switches, correlation guard	Hard halt on breach — never override
Execution agent	Submit orders to Kite, smart routing, retry, audit	Order failures logged, alerted, manual recovery
Portfolio agent	Multi-strategy allocation, Kelly sizing, rebalance	Holds the actual book of record — never deletes positions

Hosting

Layer	Choice	Why
Compute	AWS Mumbai (ap-south-1)	Same region as Kite servers — sub-50ms latency, SEBI data residency
Hot DB (ticks)	QuestDB	Fastest tick ingestion, ASOF JOIN, time-series native
Analytics DB	DuckDB + Parquet	Zero infra cost, columnar, plenty fast for our scale
Object store	S3 (Mumbai region)	Backups, model artifacts, historical Parquet
Secrets	AWS Secrets Manager	Kite tokens, API keys, never on disk in plaintext
Monitoring	Grafana + CloudWatch + Telegram	3 layers: deep dive, native, instant alert
CI/CD	GitHub Actions → AWS	Auto-deploy from <code>main</code> , manual gate for production

Tech stack (locked-in)

Concern	Tool	Cost (monthly, INR)
Live execution broker	Zerodha Kite Connect	Rs 2,000
Tick data	Kite WebSocket (3,000 instruments)	Included
Historical data download	Shoonya (Finvasia) free API	Rs 0
ML training	LightGBM + neuralforecast (TFT/LSTM)	Rs 0
LLM sentiment	Claude API (Anthropic)	~Rs 4,000
Backup local LLM	Llama 3.1 via Ollama	Rs 0
Hot DB	QuestDB (self-hosted on EC2)	included in compute
Analytics	DuckDB + Parquet	Rs 0
Cloud compute	AWS EC2 t4g.large + EBS gp3	~Rs 6,000
Storage	S3 + lifecycle to Glacier	~Rs 500
Monitoring	Grafana Cloud (free tier) → paid	Rs 0 → Rs 2,000
Telegram bot	Free	Rs 0
Domain + SSL	Cloudflare	Rs 100

Phase-by-phase monthly cost (INR):

Phase	Items	Monthly
Phase 1 (now)	Laptop + Telegram	~Rs 0
Phase 2	Cloud paper + Claude API + Shoonya	~Rs 8,000
Phase 3	Above + Kite API + AWS Mumbai + Grafana paid	~Rs 18,000
Phase 4	Above + scaled compute + monitoring	~Rs 30,000

These are pre-revenue costs. Revenue model in section 9.

Risk rules (non-negotiable)

These come from the April 12 plan and are unchanged. They are the load-bearing safety layer.

Rule	Limit	Why
Daily max loss	2% of capital	Survive any single bad day
Weekly max loss	5% of capital	Survive any bad week
Monthly max loss	10% of capital	Kill switch trigger — pause all trading
Single position max	10% of capital	Concentration risk
Max F&O leverage	3x	Survive a 30% gap-down
Max correlated positions	3 from same sector	Survive a sector blowup
Kill switch	Auto-fire at -2% daily	Hard stop. No override.
Recovery ladder after kill switch	25 / 50 / 75 / 100% over 15 days	Don't revenge-trade
Order size validation	Reject if > 2% of capital OR > 5x average position	Catch fat-fingers + bugs
Pre-trade sanity check	Reject if price > 3 SD from VWAP in last 5 min	Catch quote-stuffing / market-manip windows

The kill switch is the single most important piece of code in the system. It is tested in every CI run. It cannot be silenced by any other agent. If it fires, only manual intervention with a 24-hour cooldown can restart trading.

Compliance – the gate before we open to others

This is the section that changes everything else. Three regulatory regimes, three different obligations.

Regime	Who needs it	What it lets us do	What it costs
Personal trading	Anyone	Trade your own money via your own broker, automated or manual. No license needed.	Rs 0
SEBI algo trading registration (broker-side)	Anyone running automated orders through a broker	Register the algo strategy with the broker, who registers with the exchange. April 2026 SEBI rules tightened this. Needs fresh research.	TBD — likely Rs 0 to Rs 50K
SEBI Registered Investment Adviser (RIA)	Anyone giving paid investment advice to non-related third parties	Charge subscribers for signals, recommendations, or advisory services	Rs 5 lakh net worth + NISM XA + XB exams + ~3 month process. ~Rs 50K total fees

What this means for v6.1's roadmap

Phase	Compliance posture	Allowed
Phase 1, 2 (paper)	None needed	Paper trade, write blog posts, share own results in journal form
Phase 3 (shadow + live with 1 lot, personal)	SEBI algo registration via Kite	Trade your own money automated
Phase 4 personal scale	Same	Trade more of your own money
Phase 4+ public	RIA license required	Take subscribers, send paid signals, charge for access

Decision needed by July 2026: Do we apply for the RIA license or stay personal-only? If RIA, application starts in parallel with Phase 2 work because it takes ~3 months and blocks public launch. A separate compliance doc is the next deliverable after this one.

Threat surface (already documented in detail)

A 1,597-line analysis already exists at `docs/security/TRADEPILOT_V6_LIVE_TRADING_THREAT_ANALYSIS.md`. The categories:

Category	What's covered
Kite API security	Token theft, session hijack, rate limit abuse, key vs secret vs access token danger ranking
Financial threats	Rogue orders, flash crash, double ordering, market manipulation rules, slippage
Network	MITM, DNS hijack, SSL pinning, VPN, internet drop mid-trade
Infrastructure	SSH compromise, DDoS, DB corruption, power failure, clock sync
Quantum	Harvest-now-decrypt-later (long-term)

For v6.1 production launch, every Phase 3 item must reference its corresponding threat category and demonstrate a control. **No code goes live without a paired threat-mitigation entry.**

Revenue model (Phase 4+, contingent on RIA)

From the April 12 plan, refreshed.

Tier	Price (INR/month)	What they get	Volume target Y1
Free	Rs 0	Market data, basic signals, paper trading sandbox	10,000
Pro	Rs 999	Full signals, regime alerts, Telegram bot	500
Algo	Rs 4,999	API access, auto-execution via their own Kite, all strategies	50
Enterprise	Custom (>Rs 25K)	White-label, dedicated support, custom strategies	5

Revenue projection (Year 1 post-RIA launch):

Tier	Users	MRR	Annual
Free	10,000	Rs 0	Rs 0
Pro	500	Rs 5,00,000	Rs 60 lakh
Algo	50	Rs 2,50,000	Rs 30 lakh
Enterprise	5	Rs 1,25,000	Rs 15 lakh
Total	10,555	Rs 8.75 lakh MRR	Rs 1.05 Cr ARR

This is a conservative re-scoping. The April 12 plan said Rs 20L MRR; that assumed faster ramp. Year 1 we hit Rs 8.75L MRR. Year 2 target is Rs 25L MRR.

Risk register (SWOT-style)

Risk	Category	Mitigation
Insufficient sample at May 25 gate (no engine crosses Sharpe 1.0)	Statistical	Extend observation window 4 weeks. Do not lower the bar.
SEBI April 2026 rules disqualify our algo registration	Regulatory	Compliance doc due first week of May with full rule reading + workaround paths
Kite API outage during a signal-fire window	Technical	Smart order router with automatic retry + halt-on-stale-quote; documented incident at competitive-analysis.md:125
Streak / Sensibull copy our edge once visible	Competitive	Multi-layer alpha (Phase 2) + execution moat (sub-50ms) — they can't easily replicate
Solo founder bandwidth (Soumya also runs DevPilot + Sidewall)	Personal	Strict no-engine-changes-mid-day rule, MindPilot framework cadence, automation everywhere
ML model degrades silently (alpha decay)	ML	4-gate promotion + rolling 5-day IC tracking + auto-alert if IC turns negative
Real-money Sharpe diverges from paper Sharpe (transaction cost surprise)	Live transition	4-week shadow mode + 4-week 1-lot live before scaling
Kill switch bug	Catastrophic	Daily CI test + manual quarterly drill

Today's actionable checklist (next 4 weeks)

Concrete items between now and the May 25 gate.

Item	Owner	When	Status
Continue 6-engine observation window	Engines (auto)	Daily through 2026-05-25	Active
Watch for Fix #1 firing pattern in v5 logs	Soumya	Daily	Pending
Watch v6 vs v4 head-to-head	Soumya	Daily via auto EOD report	Pending
Read SEBI April 2026 algo trading rules	Soumya / Claude (research agent)	First week of May	Not started
Write compliance roadmap doc (the "B" option)	Claude	Week of May 5	Pending
Apply for Zerodha Kite Connect dev account	Soumya	After May 25 gate passes	Blocked on G1
Start drafting Kite execution agent code	Claude	After May 25 gate passes	Blocked on G1
Open Shoonya account (for historical data download)	Soumya	Phase 2 entry	Optional
Set up AWS Mumbai account + IAM	Soumya	Phase 3 prep	Not started
RIA license decision	Soumya	By July 2026	Open

Today's gate before doing any of this: does the May 25 statistical test pass?
Everything downstream depends on at least one engine crossing the Sharpe-1.0 / CI-positive bar.

Reference docs index

The supporting material for this roadmap, in priority order.

Doc	Use it for
<code>docs/IMPLEMENTATION_BRIEF_2026-04-27.md</code>	Statistical gate methodology, Track A spec, ML promotion gates
<code>docs/security/TRADEPILOT_V6_LIVE_TRADING_THREAT_ANALYSIS.md</code>	Every Phase 3 security control
<code>docs/research/2026-04-12_v6_master_plan.md</code>	The original v6 master plan (this doc supersedes it)
<code>docs/research/competitive-analysis.md</code>	Competitor moats, Kite outage history, SEBI rule context
<code>docs/learning/2026-04-28-eod-summary.md</code>	The v4 vs v5 RCA that motivated Fix #1 and v6
<code>docs/learning/2026-04-29-v5-vs-v6-experiment.md</code>	Tomorrow's experiment design + decision matrix
<code>docs/observation_journal.md</code>	Daily observation log through May 25
<code>docs/RETIRED_ENGINES.md</code>	Why engines were retired

Doc	Use it for
	(and why some came back)

One-line summary

v6.1 is a 36-40 week journey from today's 6-engine paper trading laptop setup to a public-facing, SEBI-compliant, multi-broker production system. We are 80% through Phase 1. The May 25 statistical gate is the next decision point. Nothing ships to live trading until at least one engine crosses Sharpe 1.0 with a positive 95% CI lower bound.